

**Summary & Objective:** An experienced data analyst with data engineering/development professional exposure and strong quantitative modeling and machine learning technics, excellent communication skills and passion for tech, looking for a sustainable career in US.

## EDUCATION

### University of Michigan - Ann Arbor

MS, Applied Statistics

Ann Arbor, MI

08/2023 - 05/2025 (expected)

### Virginia Polytechnic Institute and State University (Virginia Tech)

Double Major: B.S. Mathematics / B.S. Computer Modeling Data Analysis

Blacksburg, VA

08/2019 - 05/2023

Double Minor: B.S. Statistics / B.E. Computer Science

11/2021 - 05/2023

#### • Honors:

- **Dean's List:** Spring 2023, Fall 2022, Spring 2022, Fall 2021, Spring 2021
- **T.W. Hatcher Mathematical Scholarship:** 2022-2023
- **Stipend:** First Undergraduate Researcher, RA (06/2022 to 05/2023), TA (01/2023 - 05/2023)

## PROGRAMMING & QUALIFICATIONS

- **Data Science:** SQL, PySpark, Hadoop, Machine Learning (Logistic Regression, XGBoost, Random Forest, KNN, CNN), Statistical Analytics (A/B testing, EDA, time series, causal inference), ETL, Dashboard (Tableau /Quarto/Power BI), NLP (NLTK)
- **Programming Technologies:** Python, JAVA, C, HTML/CSS
- **Statistical Tools:** SAS (Certified Specialist), R, Excel

## PROGRAMMING & QUALIFICATIONS

### Penske Truck Leasing Co.

Data Scientist Intern in Customer Experience Team

Reading, PA

05/2024 - 07/2024

- Participated in designed and deploying the key components of customer analytics platform supported with big data (1MM+ transaction records) in **SAS** and **Python** by combing RFM analysis with K-mean clustering and ML-based churn prediction to fulfill business requirements of model centralization and improve accuracy of personalized marketing strategies (20% higher retention rate).
- Validated platform enhancements and data ingestions weekly, collaborating with senior analysts to compile performance reports.
- Teamed with senior scientists to automate a data monitoring process in Python for Penske data warehouse and develop a web-based visualization tool for business teams to identify and override/correct +15 types of data anomalies including inconsistent data types, outliers, improper data proxy usage, and feeding issues of upstream data sources.
- Provided strategic deliveries and remediation plans for self-identified data issues or ad hoc business requirements.
- Maintained the memory allocation for all **Spark** jobs and coordinated across end users for production releases; monitored the cluster/queue usage and run log of daily batches.

## PROJECT EXPERIENCE

### Aggregation Tool of Customer Analytics

Penske Truck Leasing Co.

Reading, PA

05/2024 - 07/2024

- Built an analytics platform equipped with integrated customer segmentation and churn prediction capabilities leveraging big data technologies and machine learning to deliver personalized marketing strategies from +1 million records with dimensions of transactional behaviors and demographic sources.
- Optimized the data pipeline (Python on local machine), reducing the run time by 50%, with a scalable **ETL** pipeline using Spark backed with **Hadoop** for transaction-level data ingestion, processing and feature engineering with tree-structured data as intermediate outputs for business usage.
- Implemented RFM-based customer segmentation supported by DBSCAN and K-mean clustering (with PCA functioned only working for high dimensional data) to identify key customer segments.
- Developed and trained churn prediction model utilizing supervised learning algorithms (Logistic Regression, Random Forests, and XGBoost) with approved model performance of 0.92 AUC-ROC score and reducing false positives by 18.6%.
- Established web tools and interactive dashboards in Tableau to provide stakeholders with actionable insights, including real-time visualizations of key customer metrics, rolling segment-specific trends and churn prediction results.

### Does Plant Dance: AI-enabled Monitoring and Detection of Plant Dancing

Undergraduate Researcher to Dr. Bingyu Zhao & Dr. Xinwei Deng, Virginia Tech

Blacksburg, VA

06/2022 - 05/2023

- Collaborated with an interdisciplinary research team from the Statistics Dept. and School of Plant and Environmental Sciences to develop AI-driven monitoring techniques for detecting subtle plant phenotype changes under stress conditions.
- Implemented advanced data wrangling techniques in **R** (e.g., dplyr, ggplot2) to clean, process, and visualize plant health data across 100+ Excel datasets, ensuring accurate representation of monitoring proxies.
- Developed and trained standalone ARIMA models for individual plants in treatment and control groups, applying F-tests on model parameters to identify statistically significant movement discrepancies. Results contributed to understanding plant stress responses.
- Implemented and evaluated anomaly detection algorithms (moving average, Z-score, classification models) on multivariate time series data from six plant batches, improving detection of stimulus-driven movements and refining plant stress monitoring techniques.